$\mu = 53.65/75$, $\sigma = 14$, med = 57

ECE 329 | Fields and Waves I | Summer 19

Kudeki

Exam 3

Wed, July 24, 2019 — 11 AM - 11:55

Name:	Solution.
Section:	

Please clearly PRINT your name in CAPITAL LETTERS and circle your section in the above boxes.

This is a open book exam and calculators are allowed. Please show all your work and make sure to include your reasoning for each answer. All answers should include units wherever appropriate.

Problem 1 (25 points)	
Problem 2 (25 points)	
Problem 3 (25 points)	
TOTAL (75 points)	

1. This problem has two independent parts:

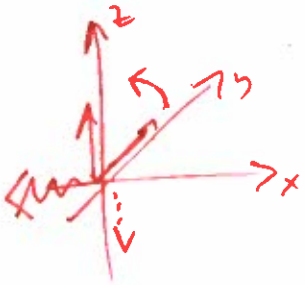
a) Consider plane TEM wave with the electric field vector

$$\mathbf{E}(\mathbf{r},t) = \cos(\omega t + \beta x)\,\hat{y} + \sin(\omega t + \beta x)\,\hat{z}\,\frac{\mathrm{V}}{\mathrm{m}}$$

propagating in free space.

i. (3 pts) Determine the corresponding phasor $\tilde{\mathbf{E}}(x)$. **Hint:** this will be a superposition of $\hat{y}$ and $\hat{z}$ directed phasors.

$$\tilde{\vec{E}}(x) = \hat{y}\,e^{+j\beta x} - j\hat{z}\,e^{+j\beta x} = (\hat{y} - j\hat{z})\,e^{+j\beta x}$$



ii. (3 pts) Determine the corresponding magnetic field intensity phasor $\tilde{\mathbf{H}}(x)$. **Hint:** make use of $\eta_o = 120\pi\ \Omega$ and the right hand rule.

$$\tilde{\vec{H}}(x) = \frac{1}{120\pi}\left(-\hat{z} - j\hat{y}\right)e^{+j\beta x}$$

iii. (3 pts) What is the propagation direction of wave field $\mathbf{E}(\mathbf{r},t)$? Tell your reason for your answer.

$-\hat{x}$ direction of course ... $\Rightarrow$ from $(\omega t + \beta x)$ with + sign!

iv. (3 pts) If $\mathbf{E}_1(\mathbf{r},t) \equiv \hat{y}(\mathbf{E}(\mathbf{r},t)\cdot\hat{y})$, how much average energy does the new wave field $\mathbf{E}_1(\mathbf{r},t)$ transport per unit time per unit area by itself in its propagation direction? Hint: we need $\langle \mathbf{E}_1(\mathbf{r},t) \times \mathbf{H}_1(\mathbf{r},t)\rangle$.

$$\bar{E}_1 = \cos(\omega t + \beta x)\,\hat{y} \Rightarrow \langle \bar{E}_1 \times \bar{H}_1 \rangle = -\hat{x}\frac{|E_1|^2}{2\eta_0} = \frac{1}{240\pi}(-\hat{x})\ \frac{W}{m^2}$$

v. (5 pts) What is the time average Poynting vector $\langle \mathbf{E}(\mathbf{r},t) \times \mathbf{H}(\mathbf{r},t)\rangle$? Hint: you can make use of the result of part (iv).

$$\langle \bar{E} \times \bar{H} \rangle = \frac{-\hat{x}}{120\pi}$$ (twice part iv)

vi. (3 pts) How is the wave field $\mathbf{E}(\mathbf{r},t)$ polarized? linear, right-circular, or left-circular? Show your work.

$\hat{y}$ leads $\hat{z}$, rotation from y to z (as shown above).

LHC !!

b) In a conducting medium $\mathbf{E} = 2\hat{x}\cos(\omega t)$ V/m at $\mathbf{r} = 0$ while conductivity $\sigma = 10^{-3}$ S/m.

i. (3 pts) What is the maximum and time averaged value of $\mathbf{J}\cdot\mathbf{E}$ at the same location?

$$\bar{J} = \sigma\bar{E}, \quad \bar{J}\cdot\bar{E} = \sigma\bar{E}\cdot\bar{E} = 4\times10^{-3}\cos^2(\omega t) \Rightarrow \bar{J}\cdot\bar{E}\big|_{max} = 4\times10^{-3}\ \frac{W}{m^3}$$

$$\langle \bar{J}\cdot\bar{E} \rangle = 2\times10^{-3}\ \frac{W}{m^3}$$

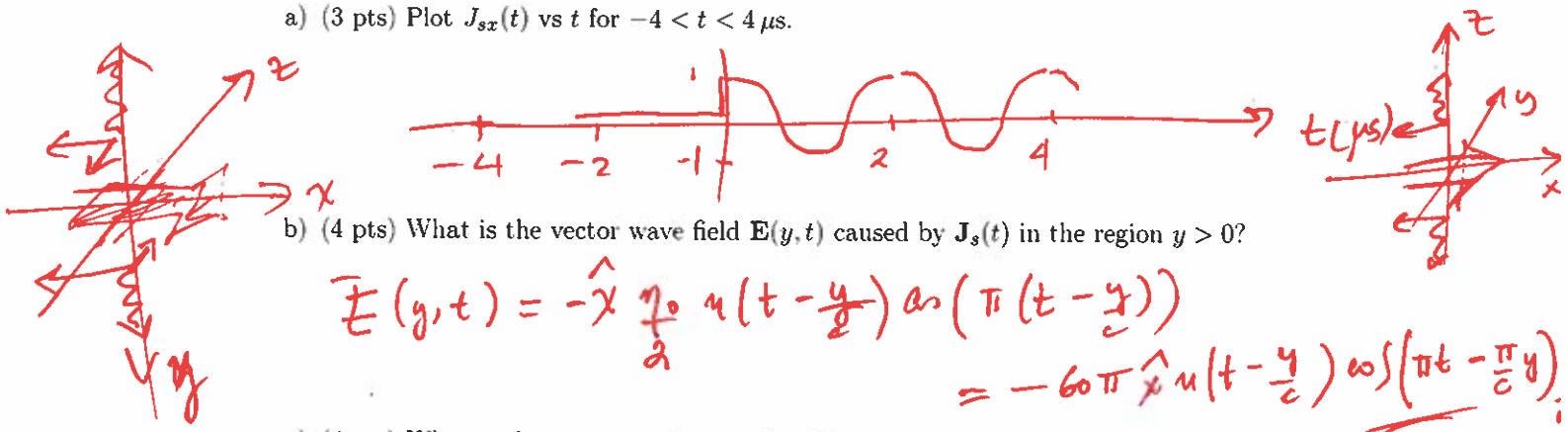
ii. (2 pts) What is the physical interpretation of $\mathbf{J}\cdot\mathbf{E}$? Hint: pay attention to the unit of $\mathbf{J}\cdot\mathbf{E}$.

$\bar{J}\cdot\bar{E}\ [=]\ \frac{W}{m^3}$ $\leftarrow$ power dissipated per unit volume

$u(t)\cos(\pi t)$ ← π rad/μs

2. Consider a time-varying surface current density $\mathbf{J}_s(t) = u(t)\cos(2\pi f t)\hat{x}$ A/m, with $f = \frac{1}{2} \times 10^6$ Hz, residing on the $y = 0$ surface embedded in free space. In this expression $u(t)$ is the unit-step function modulating the cosine signal with a frequency $\omega = 2\pi f = \pi$ rad/μs.

a) (3 pts) Plot $J_{sx}(t)$ vs t for $-4 < t < 4\,\mu$s.



b) (4 pts) What is the vector wave field $\mathbf{E}(y,t)$ caused by $\mathbf{J}_s(t)$ in the region $y > 0$?

$$\bar{E}(y,t) = -\hat{x}\,\frac{\eta_0}{2}\,u\left(t-\frac{y}{c}\right)\cos\left(\pi\left(t-\frac{y}{c}\right)\right)$$

$$= -60\pi\,\hat{x}\,u\left(t-\frac{y}{c}\right)\cos\left(\pi t - \frac{\pi}{c}y\right)$$

For $y > 0$

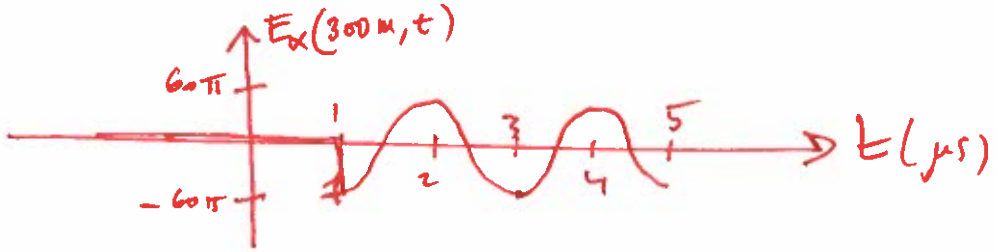
c) (4 pts) What is the vector wavefield $\mathbf{H}(y,t)$ in the region $y < 0$?

$$\bar{H}(\bar{y},t) = -\hat{z}\,\frac{1}{2}\,u\left(t+\frac{y}{c}\right)\cos\left(\pi t + \frac{\pi}{c}y\right)\ \text{A/m}\ //$$

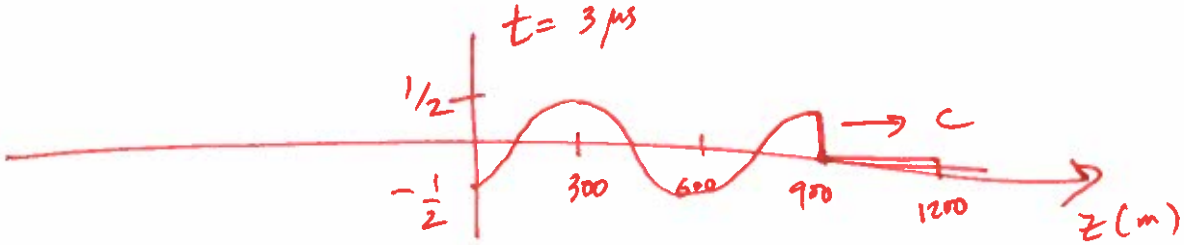
d) (3 pts) What is the wavelength λ of the modulated $\mathbf{E}(y,t)$ and $\mathbf{H}(y,t)$ waveforms?

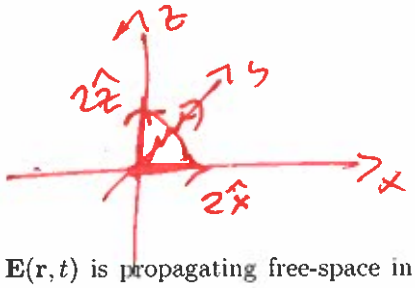
$\frac{\pi}{300} \times 600 = 2\pi \Rightarrow \lambda = 600$ m or $\lambda f = c \Rightarrow \lambda = \frac{c}{f} = \frac{300\ \text{m/μs}}{\frac{1}{2}\ 1/\mu\text{s}} = 600$ m

e) (5 pts) Plot $E_x(y,t)$ vs t at $y = 300$ m over the time interval $0 < t < 5$ μs. Label both axes carefully. **Hint:** E_x is initially zero until the wavefront arrives at 300 m.



f) (6 pts) Plot $H_z(y,t)$ vs y at $t = 3\,\mu$s over the region $0 < y < 1200$ m. Label both axes carefully. **Hint:** H_z is zero at distances where the wavefront has not yet reached.





3. This problem has two independent parts:

a) A co-sinusoidal plane TEM wave field $\mathbf{E}(\mathbf{r},t)$ is propagating free-space in $+y$ direction. The wave frequency is $f = 150$ MHz. The instantaneous electric field vector of the wave is $2\hat{x}$ V/m at time $t = 0$ at the location $(x,y,z) = (0,0,0)$, and $2\hat{z}$ V/m at $t = T/4$. where T is the wave period.

i. (4 pts) Compute the wave period T, wavelength λ, and wavenumber β.

$T = \frac{1}{150\times10^6} = \frac{1}{150}\,\mu s$; $\lambda f = c \Rightarrow \lambda = \frac{300\times10^6\ \frac{m}{s}}{150\times10^6\ \frac{1}{s}} = 2\ m$

$\beta = \frac{2\pi}{\lambda} = \pi\ \frac{rad}{m}$

ii. (4 pts) Is the wave right-circular or left-circular polarized? Explain.

LCP

iii. (4 pts) What is the phasor $\tilde{\mathbf{E}}$ expressed in terms of position y and the numerical value of β?

$\tilde{E} = 2(\hat{x} - j\hat{z})e^{-j\beta y} = 2(\hat{x} - j\hat{z})e^{-j\pi y}$ V/m.

b) Consider TEM wave propagation in $-z$ direction in some homogeneous medium with $\mu = \mu_o$, $\epsilon = 81\epsilon_o$, $\sigma = 4$ S/m — the wave frequency is $f = 1$ kHz and the wavelength is longer than the penetration depth.

$\Rightarrow$ Good conductor

i. (4 pts) Does the medium behave as an *imperfect dielectric* or else as a *good conductor* at the given wave frequency? Explain why.

$\alpha = \beta \Rightarrow \lambda = \frac{2\pi}{\alpha} > \delta = \frac{1}{\alpha}$

ii. (2 pts) What is the numerical value of the penetration depth δ — simplify without using calculator?

$\delta = \frac{1}{\sqrt{\pi f \mu \sigma}} = \frac{1}{\sqrt{\pi\, 10^3\, 4\pi\times10^{-7}\, 4}} = \frac{1}{4\pi\times10^{-2}} = \frac{25}{\pi}$ m.

iii. (2 pts) What is the numerical value of wavelength λ — simplify without using calculator?

$\lambda = 2\pi\delta = 50$ m.

iv. (2 pts) What is the numerical value of the propagation velocity $v_p = \omega/\beta$ — simplify without using calculator?

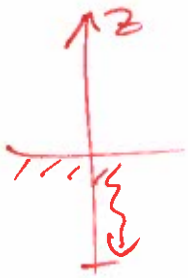
$v_p = \frac{\omega}{\beta} = \frac{2\pi\, 10^3}{\frac{2\pi}{50}} = 50\times10^3\ \frac{m}{s} = 5\times10^4\ \frac{m}{s}$.

v. (3 pts) If the wave is transporting 1 W/m^2 power density at $z = 0$ what is the power density transported at $z = -10$ m? — you can use your calculator in this part.

$\langle S\rangle\big|_{z=-10m} = e^{+2\alpha z}\langle S\rangle\big|_{z=0} = e^{-2\left(\frac{\pi}{25}\right)10}$

-10; 1

$= e^{-\frac{4}{5}\pi} \approx 0.081\ \frac{W}{m^2}$



$\mu = 58/75$, $\sigma = 7$, med $= 60$

ECE 329 | Fields and Waves I | Summer 19

Kudeki

Exam 3 - Conflict

Wed, July 24, 2019 — 9 PM - 9:55

Name:	Solution.
Section:	

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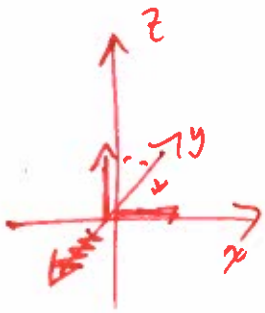
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TOTAL (75 points)	

1. This problem has two independent parts:

a) Consider plane TEM wave with the electric field vector

$$\mathbf{E}(\mathbf{r},t) = \cos(\omega t + \beta y)\,\hat{x} - \sin(\omega t + \beta y)\,\hat{z}\,\frac{\mathrm{V}}{\mathrm{m}}$$



propagating in free space.

i. (3 pts) Determine the corresponding phasor $\tilde{\mathbf{E}}(y)$. **Hint:** this will be a superposition of $\hat{x}$ and $\hat{z}$ directed phasors.

$\tilde{E}(y) = \hat{x}\,e^{j\beta y} + j\hat{z}\,e^{j\beta y} = (\hat{x} + j\hat{z})\,e^{j\beta y}$ V/m.

ii. (3 pts) Determine the corresponding magnetic field intensity phasor $\tilde{\mathbf{H}}(y)$. **Hint:** make use of $\eta_o = 120\pi\,\Omega$ and the right hand rule.

$\tilde{H}(y) = \dfrac{(\hat{z} + j(-\hat{x}))\,e^{j\beta y}}{\eta_0} = \dfrac{1}{120\pi}(\hat{z} - j\hat{x})\,e^{-j\beta y}$ A/m.

iii. (3 pts) What is the propagation direction of wave field $\mathbf{E}(\mathbf{r},t)$? Tell your reason for your answer.

$-\hat{y}$ direction

iv. (3 pts) If $\mathbf{E}_1(\mathbf{r},t) \equiv \hat{x}(\mathbf{E}(\mathbf{r},t)\cdot\hat{x})$, how much average energy does the new wave field $\mathbf{E}_1(\mathbf{r},t)$ transport per unit time per unit area by itself in its propagation direction? Hint: we need $\langle \mathbf{E}_1(\mathbf{r},t) \times \mathbf{H}_1(\mathbf{r},t)\rangle$.

$\bar{E}_1 = \cos(\omega t + \beta y)\,\hat{x} \Rightarrow \langle \bar{E}_1 \times \bar{H}_1 \rangle = -\hat{y}\,\dfrac{|E_1|^2}{2\eta_0} = \dfrac{1}{240\pi}(-\hat{y})\ \dfrac{W}{m^2}$

v. (5 pts) What is the time average Poynting vector $(\mathbf{E}(\mathbf{r},t) \times \mathbf{H}(\mathbf{r},t))$? Hint: you can make use of the result of part (iv).

$\langle \bar{E} \times \bar{H} \rangle = \dfrac{-\hat{y}}{120\pi}$ (twice part iv)

vi. (3 pts) How is the wave field $\mathbf{E}(\mathbf{r},t)$ polarized? linear, right-circular, or left-circular? Show your work.

LHC.

b) In a conducting medium $\mathbf{J} = 10^{-3}\hat{y}\cos(\omega t)$ A/m^2 at $\mathbf{r} = 0$ while the conductivity $\sigma = 10^{-3}$ S/m.

i. (3 pts) What is the maximum and time averaged value of $\mathbf{J}\cdot\mathbf{E}$ at the same location?

$\bar{E} = \dfrac{\bar{J}}{\sigma} \Rightarrow \bar{J}\cdot\bar{E} = \dfrac{\bar{J}\cdot\bar{J}}{\sigma} = \dfrac{10^{-6}\cos^2(\omega t)}{10^{-3}} = 10^{-3}\cos^2(\omega t)$

$(\bar{J}\cdot\bar{E})_{max} = 10^{-3}$ W/m^2

ii. (2 pts) What does $\mathbf{J}\cdot\mathbf{E}$ measure? Hint: pay attention to the unit of $\mathbf{J}\cdot\mathbf{E}$.

$\langle \bar{J}\cdot\bar{E} \rangle = \frac{1}{2}\,10^{-3}\ \frac{W}{m^3}$

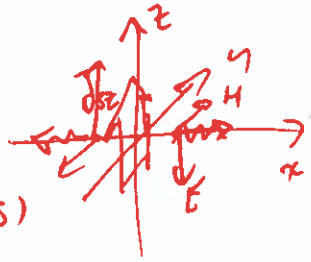
$\bar{J}\cdot\bar{E}\ [=]\ \frac{W}{m^3}$ ↪ power dissipated per unit vol.

$u(t)\cos(2\pi t)$ ← rad/μs

2. Consider a time-varying surface current density $\mathbf{J}_s(t) = u(t)\cos(2\pi f t)\hat{z}$ A/m, with $f = 10^6$ Hz, residing on the $x = 0$ surface embedded in free space. In this expression $u(t)$ is the unit-step function modulating the cosine signal with a frequency $\omega = 2\pi f = 2\pi$ rad/μs.

a) (3 pts) Plot $J_{sz}(t)$ vs t for $-3 < t < 3\,\mu$s.





b) (4 pts) What is the vector wave field $\mathbf{E}(x,t)$ caused by $\mathbf{J}_s(t)$ in the region $x > 0$?

$$\bar{E}(x,t) = -\hat{z}\frac{\eta_0}{2}u\left(t-\frac{x}{c}\right)\cos\left(2\pi\left(t-\frac{x}{c}\right)\right)$$
$$= -60\pi\,\hat{z}\,u\left(t-\frac{x}{c}\right)\cos\left(2\pi t - \frac{2\pi}{c}x\right)\ \text{V/m}$$

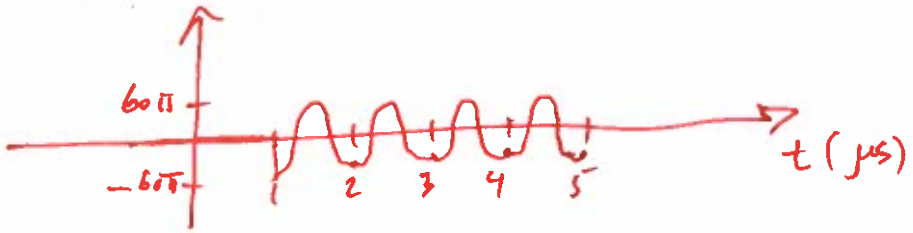
c) (4 pts) What is the vector wavefield $\mathbf{H}(x,t)$ in the region $x < 0$?

$$\bar{H}(x,t) = -\hat{y}\frac{1}{2}u\left(t+\frac{x}{c}\right)\cos\left(2\pi t + \frac{2\pi}{c}x\right)\ \text{A/m}$$

d) (3 pts) What is the wavelength λ of the modulated $\mathbf{E}(x,t)$ and $\mathbf{H}(x,t)$ waveforms?

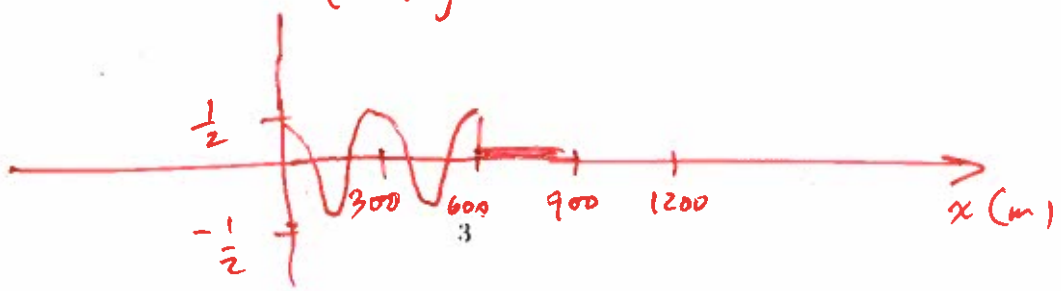
$\frac{2\pi}{300}\times 300 = 2\pi \Rightarrow \lambda = 300$ m or $\lambda f = c \Rightarrow \lambda = \frac{c}{f} = \frac{300\ \text{m/μs}}{1\ \text{1/μs}} = 300$ m

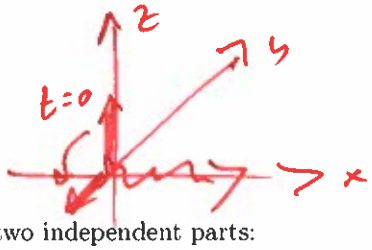
e) (5 pts) Plot $E_z(x,t)$ vs t at $x = 300$ m over the time interval $0 < t < 5$ μs. Label both axes carefully. **Hint:** E_z is initially zero until the wavefront arrives at 300 m.



f) (6 pts) Plot $H_y(x,t)$ vs x at $t = 2\,\mu$s over the region $0 < x < 1200$ m. Label both axes carefully. **Hint:** H_y is zero at distances where the wavefront has not yet reached.

$t = 2$ μs





3. This problem has two independent parts:

a) A co-sinusoidal plane TEM wave field $\mathbf{E}(\mathbf{r},t)$ is propagating free-space in $+x$ direction. The wave frequency is $f = 300$ MHz. The instantaneous electric field vector of the wave is $2\hat{z}$ V/m at time $t = 0$ at the location $(x, y, z) = (0, 0, 0)$, and $-2\hat{y}$ V/m at $t = T/4$, where T is the wave period.

i. (4 pts) Compute the wave period T, wavelength λ, and wavenumber β.

$T = \frac{1}{f} = \frac{1}{300\times10^6} = \frac{1}{300}\,\mu s$; $\lambda f = c \Rightarrow \lambda = \frac{300\times10^6\ m/s}{300\times10^6\ 1/s} = 1\,m$ //

$\beta = \frac{2\pi}{\lambda} = 2\pi\ \frac{rad}{m}$ //

ii. (4 pts) Is the wave right-circular or left-circular polarized? Explain.

RCP

iii. (4 pts) What is the phasor $\tilde{\mathbf{E}}$ expressed in terms of position x and the numerical value of β?

$2(\hat{z}+j\hat{y})\,e^{-j2\pi x}$ V/m.

b) Consider TEM wave propagation in $-z$ direction in some homogeneous medium with $\mu = \mu_o$, $\epsilon = 81\epsilon_o$, $\sigma = 4$ S/m — it is observed that the penetration depth is much longer than the wavelength.

i. (3 pts) Does the medium behave as an ~~*imperfect dielectric*~~ or else as a *good conductor* at the wave frequency? Explain why.

since $\delta \gg \lambda$

ii. (3 pts) According to (i) what inequality the wave frequency $\omega = 2\pi f$ must satisfy?

$\omega\varepsilon \gg \sigma$

iii. (2 pts) Show that $f = 10$ GHz (G=10^9) frequency satisfies the inequality in (ii).

$(2\pi\times10\times10^9)\,\frac{81}{36\pi\times10^9} = \frac{20\times81}{36} \gg 4$ // ✓ GHz

iv. (3 pts) What is the numerical value of wavelength λ in the medium if $f = 10$? — show your work prior to using calculator?

$\lambda \cong \frac{c/9}{f} = \frac{1/3\times10^8\ m/s}{10^{10}\ 1/s} = \frac{1}{3}\times10^{-2}\,m = \frac{1}{3}\,cm$

v. (2 pts) What is the numerical value of the propagation velocity $v_p = \omega/\beta$ — simplify without using calculator?

$v_p \cong \frac{1}{\sqrt{\mu\varepsilon}} = \frac{1}{\sqrt{81\epsilon_o\mu_o}} = \frac{c}{\sqrt{81}} = \frac{c}{9} = \frac{3\times10^8}{9}\ m/s$

$= \frac{1}{3}\times10^8\ m/s$